QVolution - QHackathon

Our standard problem :

$$\frac{dx(t)}{dt} = Mx(t) + b \quad \text{[First order]} \quad \text{----(1)}$$

Competition problem :

$$y'' + \omega^2 y = 0 \text{ --(2)}, \quad \omega = 1, \; y(0) = 1, \; y'(0) = 1$$

[Second order !!]

Let, $\left.\begin{aligned} x_1 &= y \\ x_2 &= y' \end{aligned}\right\}$ --(3) where a vector $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

⇒ Trying to make the competition problem look like the paper equation

Since $\omega = 1$, from equ^n (2),

$$y'' = -y$$

$$\Rightarrow \frac{dx_2}{dt} = -x_1 \text{ -----(4)}$$

Hm. Also, $\frac{dx_1}{dt} = y' = x_2$ ----(5)

So, $\frac{dX}{dt} = \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix}$ ----(6) (?i)

We need some relation between

$$X \text{ and } \frac{dX}{dt}$$

i.e. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\begin{bmatrix} x_2 \\ -x_1 \end{bmatrix}$

<u>Trial 1 for M</u>

Since places are swapped M should look like the bit flip operator i.e. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

but there's a minus i.e. a phase change for the second element. So, should be

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \end{bmatrix}$$

or, $$MX = \frac{dX}{dt}$$

then so our $b = 0$ (Great !!)

Rewriting $$\frac{dX}{dt} = MX \text{ where } M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \text{----(7)}$$

So, M should be unitary if we want to work on quantum computers. Also, I noticed our M looks quite similar to this matrix

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

which is matrix form of the Pauli Y operator.

Now, we see that

$$iY = +i\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i^2 \\ i^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = M \quad \text{---(8)}$$

Thus, M should be unitary since Y is!

Also our solution to equ^n (7) should be of the form

$$X(t) = \exp(Mt)\,X(0) \quad \text{----(9)}$$

If we can make an operator U such that

$$U|X(0)\rangle = \exp(Mt)\,|X(0)\rangle \quad \text{----(10)}$$

Our job is almost done I think!

They used the Taylor expansion to model the exponential (term) (oh sorry) terms

$$x(t) \approx \sum_{m=0}^{k} \frac{(Mt)^m}{m!} x(0) + \sum_{n=1}^{k} \frac{M^{n-1} t^n}{n!} b \quad \text{---} (11)$$

for the terms

$$\bar{x}(t) = e^{Mt} x(0) + (e^{Mt} - I) M^{-1} b$$

But in our case $b = 0$! So,

We only need the first term !!

i.e.

$$x(t) \approx \sum_{m=0}^{k} \frac{(Mt)^m}{m!} x(0) \quad \text{---} (12)$$

Oh! Since $M = iY$

$M^0 = I$, $M^1 = iY$, $M^2 = i^2 Y^2 = -I$,

$m=0$ $m=1$ $m=2$

$M^3 = (iY)^3 = -iY$ $M^4 = (iY)^4 = I$ $M^5 = M^4 M = iY$

$m=3$ $m=4$ $m=5$

So, the terms show a cyclic order reducing our workload.

$m = 0, 1, 2, 3$ and then repeats!

So repeats every 4 steps.

From equ^n (12),

$$x(t) \approx \left(I \cdot \frac{t^0}{0!} + iY \frac{t^1}{1!} + (-I)\frac{t^2}{2!} + (-iY)\frac{t^3}{3!} + I \cdot \frac{t^4}{4!} + iY \frac{t^5}{5!} + (-I)\frac{t^6}{6!} + (-iY)\frac{t^7}{7!} + \dots \right) x(0)$$

Oh! I see a pattern. We can group the common terms! (those who have (iY) and those who don't)

$$x(t) \approx \left[\underbrace{\left(\frac{t^0}{0!} - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \dots \right)}_{\cos t} + \underbrace{\left(\frac{t^1}{1!} - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \dots \right)}_{\sin t} \{iY\} \right] \cdot x(0)$$

$$x(t) \approx (\cos t + iY \sin t)\, x(0)$$

Damn! This looks like the R_y gate.

In R_Y gate,

$$RY(\theta) = \exp\left(-i\frac{\theta}{2}Y\right) = \cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\cdot Y$$

$$= \cos\frac{\theta}{2} - iY\sin\frac{\theta}{2}$$

Looks familian !

Our evolution was,

$$\cos(t) + iY\sin(t) = \exp(itY)$$

Well our evolution correspond to an inverse rotation where $\theta = 2t$.

I don't really know how that helps ...

Comparing our equ^n (13) with equ^n (47) of paper

$$C = \sqrt{\cos^2 t + \sin^2 t} = 1$$

$$D = 0$$

Thus, Operations $U_0 = I$, $U_1 = iY$

From equ^n (49) of paper, $V_{s_1} = \frac{1}{1}\begin{bmatrix} \cos\theta & N \\ \sin\theta & N \end{bmatrix}$

where Ns are arbitrary elements that make V_{s_1} unitary.

From equ^n (10) of paper,

$$V = \frac{1}{N}\begin{pmatrix} C & D \\ D & -C \end{pmatrix} = \frac{1}{N}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

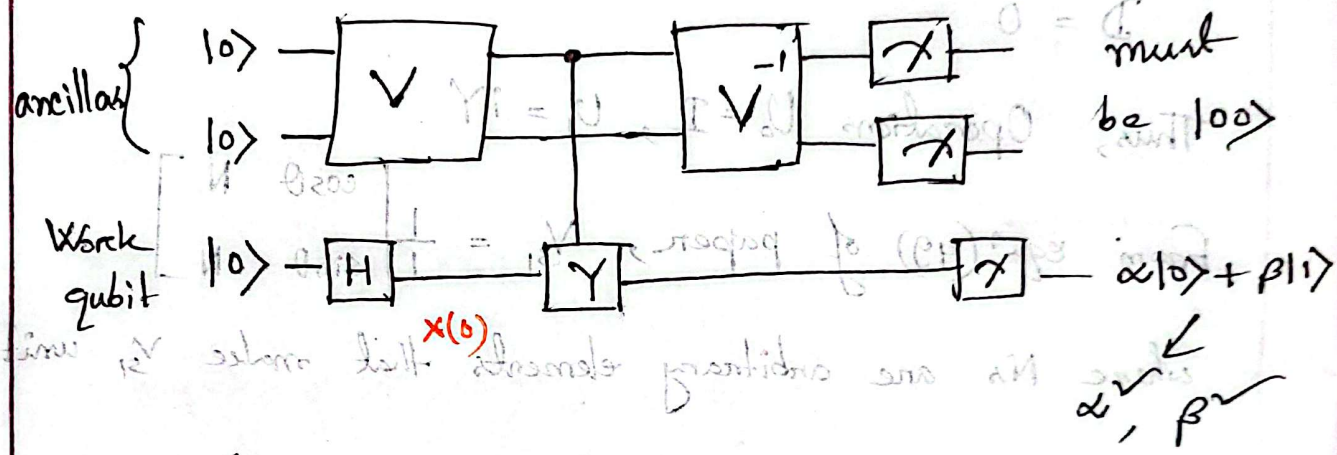
where, N is the dimension of the vector = 2

Number of work qubits $= \log_2 N = \log_2 2 = 1$

If we take $k = 4$, then $T = \log_2(k+1) = \log_2(5) \approx 3$

So, needed number of ancillas $= T+1 = 4$

Might not need the



1. Initialize :

Work qubit : $|0\rangle$:, apply H to create initial state vector $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$ i.e. $\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$ that says $y=1$, $y'=1$

Ancillas : $|00\rangle$

2. The V box : (inplace_prepare_state) (Encoding)

M^k repeats $(-1, i, -1, -i)$ scalars and so the amplitudes

$$\begin{bmatrix} \frac{t^0}{0!} \\ i\frac{t^1}{1!} \\ -\frac{t^2}{2!} \\ \vdots \end{bmatrix}$$

3. Entanglement :

Since we input the phases i.e. scalars beforehand we just need to apply :

$k \to 0$	:	I	since $M^0 = I$
$k \to 1$	:	Y	$M^1 = iY$
$k \to 2$	:	I	$M^2 = -I$
$k \to 3$	:	Y	$M^3 = -iY$

$(1 \to 01,\ 3 \to 11)$ So we apply Y only when the first ancilla qubit is 1

4. Decoding :

In claring there should be a function for finding V^{-1}. Since we have already defined a function in step 2 we just use "adjoint" to find and implement V^{-1}.

Lastly measuring find the probabilities of finding 0 and 1.

In my formulation : if $|0\rangle$ — position y.
$|1\rangle$ — velocity y'

$\alpha|0\rangle + \beta|1\rangle$

For a harmonic oscillator given $\omega = 1$, if $m = 1$ the total energy should be conserved i.e. stays the same as the initial condition.

$$y(0) = 1 \quad y'(0) = 1$$

So, $E_{H.O.} = P.E. + K.E. = \frac{1}{2} y(0)^2 + \frac{1}{2} y'(0)^2$

$$= \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

Hence, $P.E. = \alpha^2 \times E_{H.O.} = \alpha^2$

$K.E. = \beta^2 \times E_{H.O.} = \beta^2$

<u>Sanity Check</u>: Make sure to check whether the final state of the ancillas are $|00\rangle$ after doing all these operations.